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Course: Spatial Data Science

Semester Project Abstract

Project Title:

Geospatial Identification of Rooftop Solar Energy Potential Using Building Footprints and Solar Radiation Data: A Case Study of San Diego, California

Abstract

Urban areas are experiencing increasing energy demand due to population growth and rapid development. At the same time, there is a growing need to transition toward cleaner and more sustainable energy sources. Rooftop solar energy offers an effective solution for generating renewable electricity within cities without requiring additional land. However, identifying which buildings are suitable for solar panel installation across a large urban area can be difficult. Factors such as rooftop size, spatial distribution of buildings, and sunlight availability influence the potential for rooftop solar systems. Because of these challenges, spatial data analysis can be used to better understand where solar energy installation would be most effective.

This project focuses on identifying rooftops that have strong potential for solar energy generation in San Diego, California. The main research question guiding this study is: **How can spatial data analysis be used to identify rooftops that are suitable for solar panel installation in San Diego?** Understanding rooftop solar potential can help support renewable energy planning and highlight areas where urban solar adoption may be most beneficial.

The proposed methodology uses openly available geospatial datasets and Python-based spatial analysis techniques. Building footprint data from the Microsoft US Building Footprints dataset will be used to represent rooftop structures across the city. The San Diego city boundary dataset will define the study area and ensure that the analysis focuses only on buildings within the city. Solar radiation data from the National Renewable Energy Laboratory (NREL) will be used to estimate the amount of solar energy available in different parts of the city. Using Python libraries such as GeoPandas, Rasterio, and Shapely, building footprints will be processed to calculate rooftop area and spatial characteristics. These rooftops will then be combined with solar radiation data to estimate potential solar exposure for each building.

The main objectives of this project are to identify rooftop structures within the study area, evaluate rooftop characteristics such as size and distribution, and integrate solar radiation data to estimate rooftop solar suitability. The final outcome of the study will be a spatial map that highlights buildings with high potential for solar energy installation.

The expected results of this research include identifying areas within San Diego where rooftop solar systems could be most effective. The project will demonstrate how spatial data science and open geospatial datasets can support sustainable urban energy planning and provide insights for future renewable energy initiatives.